

A Small Theorem and the Discipline of Real Analysis

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1 A deceptively simple statement

Consider the following elementary statement about two real numbers x and y :

$$x = y \iff \forall \varepsilon > 0, |x - y| < \varepsilon.$$

At first sight, this hardly looks like a theorem that should require any care. If $x = y$, then their distance is zero, and zero is smaller than every positive number. Conversely, if the distance between x and y is smaller than every positive number, then surely that distance must be zero.

The mathematical content is indeed simple. Yet this small problem is an excellent example of what rigorous mathematics, and real analysis in particular, demands. The difficulty does not lie in computation. It lies in organizing the logic correctly: distinguishing hypotheses from conclusions, proving both directions of an equivalence, handling quantifiers accurately, and negating statements without changing their meaning.

In this sense, the problem is much more representative of “real mathematics” than its elementary appearance suggests.

2 The theorem

We want to prove

$$x = y \iff \forall \varepsilon > 0, |x - y| < \varepsilon.$$

Because the statement is an equivalence, there are two implications to prove:

$$x = y \implies \forall \varepsilon > 0, |x - y| < \varepsilon,$$

and

$$(\forall \varepsilon > 0, |x - y| < \varepsilon) \implies x = y.$$

Keeping these two directions explicitly separate is already an important proof-writing habit. An “if and only if” statement is not a single vague assertion of similarity; it is the conjunction of two implications.

3 The forward implication

Assume

$$x = y.$$

Then

$$x - y = 0,$$

and therefore

$$|x - y| = 0.$$

Now let $\varepsilon > 0$ be arbitrary. Since

$$0 < \varepsilon,$$

we obtain

$$|x - y| = 0 < \varepsilon.$$

Because $\varepsilon > 0$ was arbitrary, it follows that

$$\forall \varepsilon > 0, |x - y| < \varepsilon.$$

This direction is completely straightforward. Nevertheless, even here the quantifier matters. We do not merely prove the inequality for one convenient value of ε . We take an arbitrary positive ε , prove the required inequality, and only then conclude that it holds for every positive ε .

4 The reverse implication

Now assume

$$\forall \varepsilon > 0, |x - y| < \varepsilon.$$

We want to prove

$$x = y.$$

A convenient method is proof by contradiction. Suppose instead that

$$x \neq y.$$

Then

$$|x - y| > 0.$$

This gives us a particularly useful positive number:

$$\varepsilon = |x - y|.$$

Because $x \neq y$, this choice satisfies $\varepsilon > 0$, so it is allowed by the hypothesis. The hypothesis therefore implies

$$|x - y| < \varepsilon.$$

Substituting $\varepsilon = |x - y|$, we obtain

$$|x - y| < |x - y|,$$

which is impossible.

Hence the assumption $x \neq y$ must be false. Therefore

$$x = y.$$

Thus both implications hold, and so

$$x = y \iff \forall \varepsilon > 0, |x - y| < \varepsilon.$$

5 Where the real difficulty lies

Nothing in the proof requires advanced algebra or sophisticated machinery. The essential challenge is logical bookkeeping.

Suppose we are trying to prove an implication

$$P \implies Q.$$

A direct proof begins by assuming P and deriving Q . A proof by contradiction begins by assuming P together with the negation of Q , and then deriving an impossibility.

This distinction is easy to state and surprisingly easy to violate in practice. One can accidentally negate the hypothesis instead of the conclusion, or begin with a statement that is related to the theorem without actually constituting a proof of the desired implication.

The present theorem illustrates this sharply. In the reverse direction, the hypothesis is

$$\forall \varepsilon > 0, |x - y| < \varepsilon.$$

Its negation is

$$\exists \varepsilon > 0 \text{ such that } |x - y| \geq \varepsilon.$$

If one starts by assuming this existential statement, one has negated the hypothesis itself. That may be useful in proving a contrapositive, but it is not the correct starting point for a contradiction proof of the original implication.

The contradiction proof instead keeps the universal hypothesis and negates the desired conclusion:

$$\forall \varepsilon > 0, |x - y| < \varepsilon, \quad x \neq y.$$

From these two assumptions, the contradiction follows.

6 Quantifiers are mathematical structure

One of the central lessons is that quantifiers are not decorative notation. They determine the logical structure of a statement.

Compare

$$\forall \varepsilon > 0, |x - y| < \varepsilon$$

with

$$\exists \varepsilon > 0, |x - y| < \varepsilon.$$

The second statement is extremely weak. For any fixed real numbers x and y , one can simply choose a sufficiently large ε , for example

$$\varepsilon = |x - y| + 1,$$

and then

$$|x - y| < \varepsilon.$$

So the existential statement tells us essentially nothing about whether x and y are equal.

The universal statement is radically stronger. It says that no matter how small a positive tolerance we choose, the distance $|x - y|$ is smaller still. The only nonnegative real number with this property is zero.

This is the real mathematical content of the theorem.

7 Negating quantified statements

The problem also gives a useful exercise in negation.

The negation of

$$\forall \varepsilon > 0, |x - y| < \varepsilon$$

is not merely

$$|x - y| \geq \varepsilon.$$

The variable ε must still be quantified. The correct negation is

$$\exists \varepsilon > 0 \text{ such that } |x - y| \geq \varepsilon.$$

Symbolically,

$$\neg (\forall \varepsilon > 0, |x - y| < \varepsilon)$$

is equivalent to

$$\exists \varepsilon > 0 \text{ such that } |x - y| \geq \varepsilon.$$

Two changes occur simultaneously:

$$\forall \longleftrightarrow \exists,$$

and

$$< \longleftrightarrow \geq.$$

This pattern appears constantly in analysis, especially in the definitions of limits and continuity.

For example, the statement

$$\forall \varepsilon > 0, \exists \delta > 0, \dots$$

has a negation beginning with

$$\exists \varepsilon > 0, \forall \delta > 0, \dots.$$

A mistake in the order or type of quantifiers can completely change the theorem.

8 The hidden lemma

The theorem can be reduced to a very general elementary fact:

$$r \geq 0 \quad \text{and} \quad \forall \varepsilon > 0, r < \varepsilon \quad \implies \quad r = 0.$$

Indeed, simply take

$$r = |x - y|.$$

If $r > 0$, then we may choose

$$\varepsilon = r.$$

The assumption $r < \varepsilon$ would then become

$$r < r,$$

which is impossible.

Therefore $r = 0$.

This tiny lemma occurs repeatedly throughout analysis. Often one proves an inequality depending on an arbitrary positive parameter, such as

$$|x - y| \leq C\varepsilon$$

for every $\varepsilon > 0$, where $C > 0$ is fixed. Since ε can be taken arbitrarily small, one concludes

$$|x - y| = 0,$$

and hence

$$x = y.$$

The same idea appears in uniqueness proofs, limit arguments, approximation theorems, metric spaces, and many other settings.

9 Why this is characteristic of real analysis

Real analysis is full of statements whose intuitive content is simple but whose precise formulation contains several nested logical operations.

A definition of a limit, for example, has the form

$$\forall \varepsilon > 0, \exists \delta > 0$$

followed by an implication involving inequalities.

A convergence proof may require choosing one quantity in terms of another. A non-convergence proof requires negating the entire quantified statement correctly. A uniqueness proof may depend on showing that a nonnegative number is smaller than every positive number. None of these tasks necessarily involves difficult computation.

The challenge is precision.

This is one of the major transitions from computational mathematics to proof-based mathematics. In elementary calculus, one may often know what an expression “ought” to do and then perform a calculation. In analysis, the informal intuition is only the starting point. The proof must preserve the exact logical content of the theorem at every step.

The small theorem

$$x = y \iff \forall \varepsilon > 0, |x - y| < \varepsilon$$

is therefore an unusually good training example. It is simple enough that no technical machinery obscures the logic. Every mistake is exposed as a mistake in reasoning itself.

10 A useful proof-writing discipline

For elementary analysis proofs, it is often helpful to translate the theorem into a logical skeleton before doing any mathematics.

For an implication

$$P \implies Q,$$

write explicitly:

- Hypothesis: P .
- Goal: Q .

For an equivalence

$$P \iff Q,$$

split it immediately into

$$P \implies Q$$

and

$$Q \implies P.$$

If using contradiction to prove

$$P \implies Q,$$

write the assumptions

$$P \quad \text{and} \quad \neg Q$$

before proceeding.

If quantified statements occur, write their negations explicitly rather than relying on intuition.

These habits can feel overly formal for very small problems. That is precisely why small problems are a good place to practice them. Once the arguments become longer and contain several nested quantifiers, the same discipline becomes indispensable.

11 Conclusion

The theorem

$$x = y \iff \forall \varepsilon > 0, |x - y| < \varepsilon$$

contains almost no computational difficulty. Yet proving it carefully requires several fundamental habits of rigorous mathematics: separating the two directions of an equivalence, tracking hypotheses and conclusions, handling universal and existential quantifiers, negating statements correctly, and choosing a useful value of a quantified variable.

Its simplicity makes these logical features unusually visible.

This is one reason elementary real analysis can be intellectually demanding even when the underlying formulas are familiar. The subject trains not merely calculation, but exact reasoning. A proof can be mathematically close to the truth and still fail because its assumptions and conclusions are arranged incorrectly.

That distinction—between an intuitively correct idea and a logically valid proof—is one of the central lessons of proof-based mathematics.